

FINAL PROJECT – MACHINE LEARNING

BINUS University

Academic Year: 2025/2026

Faculty / Department: School of Computer Science / Computer Science Department

Course Code – Course: COMP6577001 / Machine Learning

Class: All Classes

Please submit the project on time according to the exam schedule

The penalty for CHEATING is DROP OUT!

Purpose

The final project provides students with an opportunity to integrate theoretical understanding and practical skills in Machine Learning. Students are required to design, implement, deploy, and evaluate a machine learning solution that addresses a real-world problem.

This project emphasizes an **end-to-end classical machine learning pipeline**, including data understanding, model development, system deployment, and evaluation through **real user interaction**.

Learning Outcomes (LOs)

Upon successful completion of this project, students will be able to:

- **LO1:** Explain fundamental machine learning concepts and challenges.
- **LO2:** Analyze datasets and apply regression, classification, or clustering techniques appropriately.
- **LO3:** Implement, deploy, and evaluate a machine learning system based on real user interaction and feedback.

Mapping of Deliverables to Learning Outcomes

Deliverable	Aligned Learning Outcomes
Machine Learning Model & Source Code	LO1, LO2
Deployed Application	LO2, LO3
Report (PPT)	LO1, LO2, LO3

Demo Video	LO3
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Suggested Project Themes

- Regression-based predictive systems
- Classification-based decision support systems
- Recommender systems with user interaction
- Clustering-based user segmentation tools
- End-to-end machine learning applications with user feedback loop

Important Note

Only **classical machine learning methods** are allowed (linear models, tree-based models, SVM, clustering).

Deep learning approaches are strictly not allowed in this course.

Technical Notes

- Programming Language: Python
 - Libraries: NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn
 - Deployment Frameworks (Mandatory): Flask, FastAPI, or Streamlit
 - Version Control: Git (mandatory)
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Deployment Requirement (Mandatory)

- The machine learning model **must be deployed** as a usable application.
- Deployment may be web-based or API-based.
- The deployed system must:
 - accept user input,
 - execute the trained model,
 - produce clear and interpretable output.
- Accessibility may be via:
 - local demo,
 - shared access link,

- or recorded live demonstration.

Real User Testing Requirement (Mandatory)

- Minimum **5 real users**, not part of the project team.
- Students must define:
 - anonymized user background,
 - usage scenario,
 - usability and usefulness feedback.
- Collect **quantitative** (rating/Likert) and **qualitative** feedback.
- Results **must be analyzed and presented in the PPT**.

Project Proposal (Checkpoint)

Submission: Week 6 (1–2 pages)

- Project title and problem statement
- Dataset source and description
- Selected ML methods and baseline
- Planned evaluation metrics
- Deployment and user testing plan
- References (min. 2–3)

Deliverables & Weights

Component	Weight
Machine Learning Model & Source Code	35%
Deployed Application	25%
Report (PPT, incl. User Testing)	30%
Demo Video	10%
Total	100%

Report Format Requirement (Mandatory)

- Format: **PPT / PPTX**
 - Length: **15–25 slides** (excluding cover & appendix)
 - Language: English (recommended) or Indonesian (consistent)
 - Slides must be **self-explanatory**
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Required PPT Structure

1. Cover
 2. Problem Statement & Motivation
 3. Dataset Description
 4. Exploratory Data Analysis (EDA)
 5. ML Problem Formulation
 6. Model Selection & Rationale
 7. Training & Validation Setup
 8. Evaluation Metrics & Results
 9. Deployment Architecture
 10. Application Screenshots
 11. User Testing Design
 12. User Testing Results
 13. Analysis & Discussion
 14. Limitations
 15. Conclusion & Future Work
 16. Appendix (optional)
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Evaluation Rubric (with Grade Bands)

Grade Band Definition

- **Excellent (86–100):** Complete, accurate, deep understanding
- **Good (71–85):** Correct with minor weaknesses

- **Fair (51–70):** Superficial or inconsistent
- **Poor (0–50):** Does not meet minimum requirements

1. Machine Learning Model & Source Code (35%)

Criteria	Weight	Excellent	Good	Fair	Poor
Problem formulation & ML understanding	10%	Correct and well-justified	Mostly correct	Weak framing	Incorrect
Dataset analysis & EDA	10%	Deep and insightful	Adequate	Shallow	Missing
Model selection & training	10%	Appropriate with rationale	Appropriate	Weak choice	Incorrect
Evaluation & metrics	5%	Correct and well-analyzed	Correct, minimal analysis	Weak metrics	Missing

2. Deployed Application (25%)

Criteria	Weight	Excellent	Good	Fair	Poor
Deployment completeness	10%	Fully deployed & accessible	Limited access	Unclear	Not deployed
Functionality & stability	10%	Stable, error-free	Minor issues	Frequent errors	Not functional
Usability	5%	Easy for non-technical users	Acceptable	Confusing	Unusable

3. Report (PPT, incl. User Testing) (30%)

Criteria	Weight	Excellent	Good	Fair	Poor
Structure & storytelling	6%	Logical and persuasive	Logical	Weak flow	Disorganized
Clarity of ML reasoning	8%	Clear and well-justified	Mostly clear	Superficial	Unclear
EDA & result visualization	6%	Informative with interpretation	Adequate	No interpretation	Irrelevant
Deployment explanation	5%	Clear architecture & flow	Partial	Shallow	Missing
User testing design & analysis	5%	Structured and reflective	Descriptive	Minimal	Missing

4. Demo Video (10%)

Criteria	Weight	Excellent	Good	Fair	Poor
Deployment demonstration	5%	Clear, real execution	Partial demo	Incomplete	Missing
Explanation clarity	5%	Clear and professional	Understandable	Hard to follow	No explanation

Hard Constraints

- No PPT → **final score = 0**
 - No deployment → **deployment score = 0**
 - No user testing in PPT → **maximum final score = 70**
 - Notebook-only project → **does not meet requirements**
 - Deep learning used → **out of course scope**
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Dataset Policy

- Use legal and ethical datasets.
 - Document preprocessing steps.
 - Disclose dataset limitations.
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Ethics & Responsible AI

- Avoid bias and discrimination.
 - Ensure transparency where possible.
 - Protect user privacy.
 - Avoid harmful applications.
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Submission Guidelines

- Submission via **TEAMS**:
 - GitHub repository,
 - PPT report,
 - demo video link.
 - **Deadline**: Week 14, Sunday, 23:59 WIB
 - **Late submission**: –10% per day
 - **Plagiarism**: results in failing grade.
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Verified by Department

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